INVENTORY & REORDER ALERTS
+ TEAM DASHBOARD

Power Platform Capstone Project — Submission Documentation

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Date	March 29, 2026
Platform	Microsoft Power Platform
Tools Used	Power Apps, Power Automate, Power BI, Dataverse

1. Project Overview

1.1 Business Problem

Organizations that rely on consumable supplies such as toner cartridges, LAN cables, and paper reams often struggle to maintain adequate stock levels. Without an automated tracking system, stockouts go unnoticed until they cause operational disruption.

This project solves that problem by building a fully automated Inventory & Reorder Alerts system using Microsoft Power Platform — enabling real-time stock visibility, automated low-stock alerts, and a management dashboard.

1.2 Project Specification

Topic :**Inventory & Reorder Alerts + Team Dashboard**

Business problem

Track consumables (e.g., toner, LAN cables) and get alerts when stock falls below threshold.

Core learning goals

Calculated columns, scheduled flows, low-stock alerts, compact BI dashboard.

Scope (MVP)

· Tables: Items (Name, Category, OnHand, ReorderPoint, Supplier), Movements (Item, Qty±, Date, Reason)

· App: Receive/issue stock; quick add; prevent negative OnHand

· Flow: Nightly check → low-stock email/Teams card with “Create PO” link

· Analytics: Stock by category; weeks-on-hand; issues by department

· Stretch: Auto-create draft purchase request on threshold breach

· Acceptance: OnHand updates correctly; alert fires only once until replenished

Figure 1: Project specification — Topic, business problem, learning goals and scope (MVP)

1.3 Core Learning Goals

- Calculated columns in Dataverse for derived stock status
- Scheduled Power Automate flows for nightly low-stock checks
- Automated email/Teams alerts with reorder action links
- Compact Power BI dashboard for stock analytics

1.4 Acceptance Criteria

- OnHand quantity updates correctly on every stock movement (receive or issue)
- Low-stock alert fires only once per item until it is replenished (no duplicate alerts)
- Stock Manager app prevents negative OnHand quantities
- Dashboard reflects real-time stock status across all categories

2. Data Model — Dataverse Tables

Two Dataverse tables were designed and implemented to support the inventory system:

2.1 Items Table

The Items table is the master record for all tracked consumables. Each row contains:

- Name — display name of the consumable item
- Category — grouping (Cables, Paper, Toner, Other)
- OnHand — current quantity in stock (numeric)
- ReorderPoint — minimum threshold; alerts fire below this value
- Supplier — vendor name for purchase orders
- AlertSent — Yes/No flag preventing duplicate alert emails

2.2 Movements Table

The Movements table records every stock transaction (receive or issue), providing a full audit trail:

- Item — lookup to the Items table
- Qty+ / Qty- — quantity received or issued
- Reason — notes or context for the movement

2.3 Items Table — Screenshot

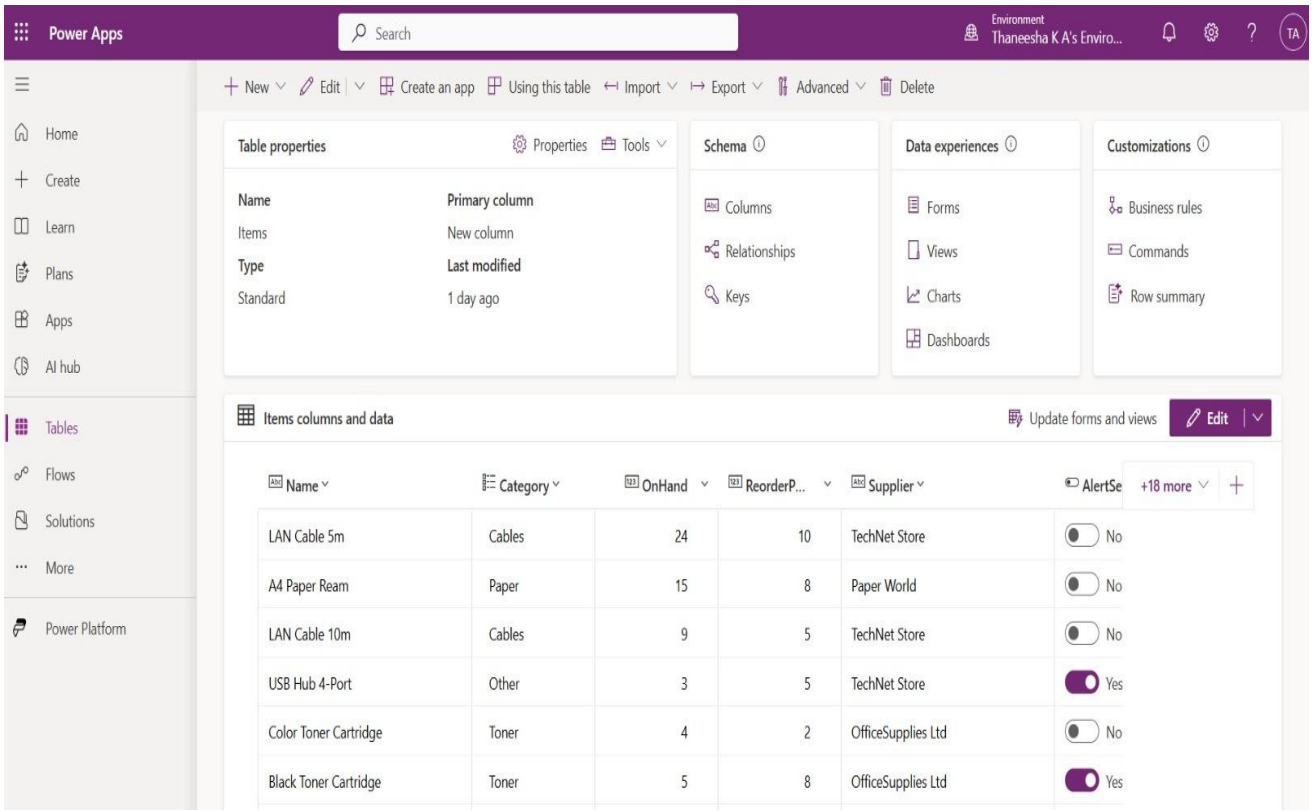


Figure 2: Items table in Power Apps Dataverse — showing Name, Category, OnHand, ReorderPoint, Supplier and AlertSent columns with live data

As shown above, the table contains 6 items across 4 categories. Items like "USB Hub 4-Port" and "Black Toner Cartridge" have AlertSent = Yes, indicating a low-stock alert has been triggered and is pending replenishment.

3. Power Apps — Stock Manager Application

A Canvas App called Stock Manager was built to allow staff to receive stock, issue stock, and view current inventory levels — all from a single mobile-friendly interface.

3.1 Main Screen — Inventory List

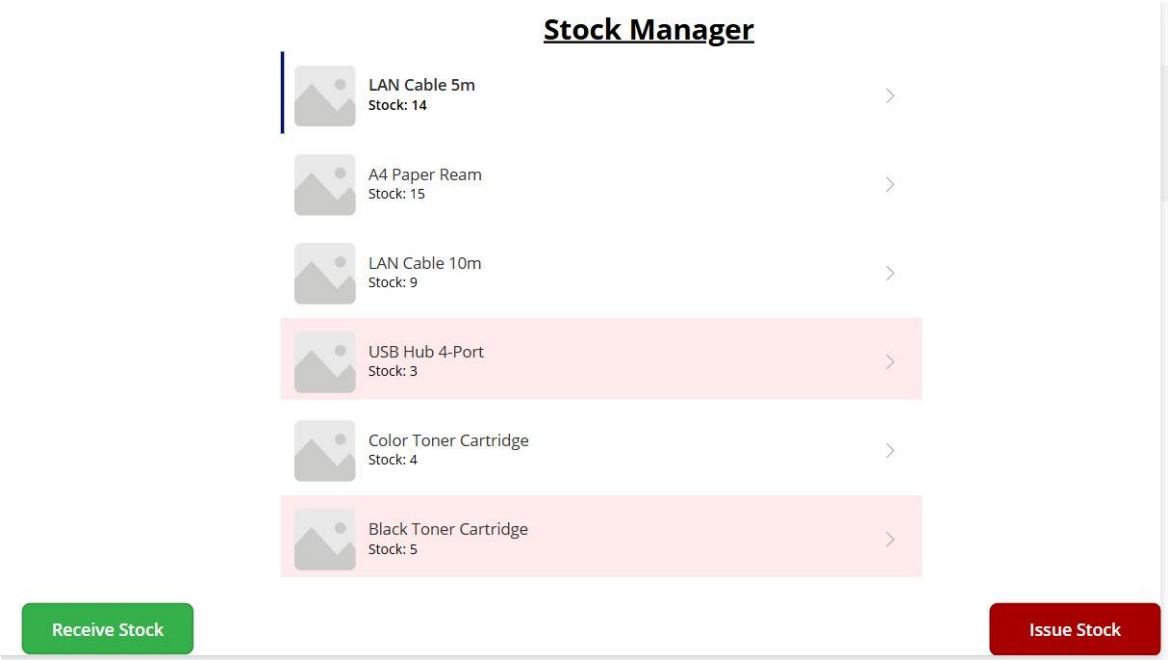


Figure 3: Stock Manager main screen — items list with low-stock items highlighted in red/pink background

Key features of the main screen:

- All items displayed with current Stock quantity
- Items below reorder point shown with a red/pink background highlight — instant visual warning
- Navigation buttons: "Receive Stock" (green) and "Issue Stock" (red) for quick access
- Tap any item to drill into details and perform stock movements

3.2 Receive / Issue Stock Screen

Figure 4: Receive/Issue Stock screen — transaction type dropdown and quantity input with Submit button

The transaction screen enforces key business rules:

- Dropdown to select transaction type: Received or Issued
- Quantity input field with numeric validation
- Submit button updates the OnHand value in Dataverse and creates a record in the Movements table
- The app prevents negative OnHand — an issue transaction is blocked if it would reduce stock below zero

4. Power Automate — Automated Alert Flows

Two Power Automate cloud flows were built to automate the alert lifecycle: one to send nightly low-stock alerts, and one to reset the alert flag when stock is replenished.

4.1 Flows Overview

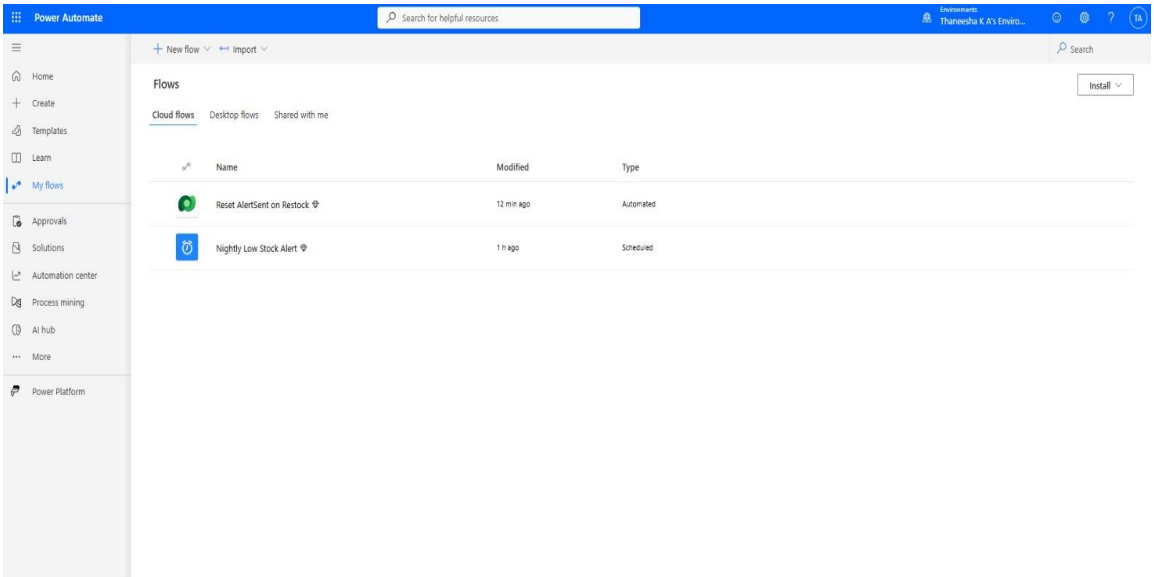


Figure 5: Power Automate — My Flows screen showing both cloud flows

Flow Name	Type	Purpose
Nightly Low Stock Alert	Scheduled	Runs nightly; queries items where OnHand < ReorderPoint and AlertSent = No; sends email alert and sets AlertSent = Yes
Reset AlertSent on Restock	Automated	Triggers when an Item row is modified; if OnHand >= ReorderPoint, resets AlertSent = No so next nightly run can alert again if needed

4.2 Nightly Low Stock Alert Email

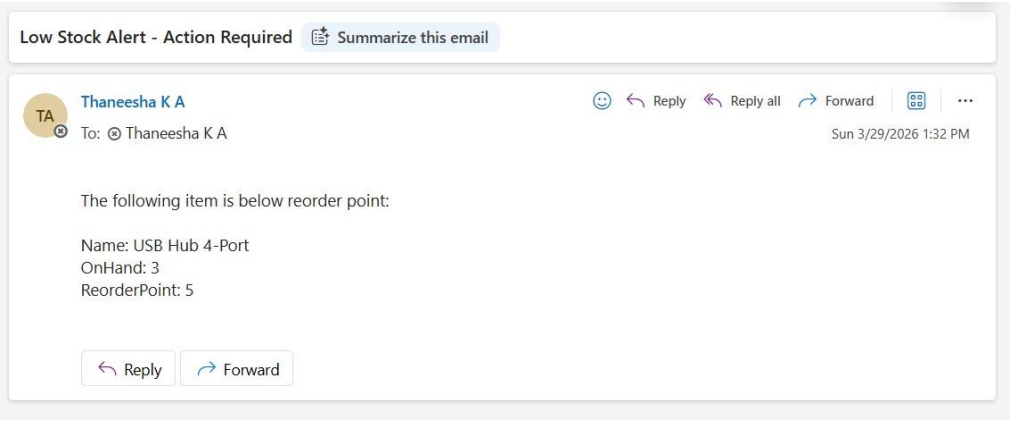


Figure 6: Email received — "Low Stock Alert - Action Required" showing item name, OnHand and ReorderPoint

The email alert clearly communicates:

- Subject: "Low Stock Alert - Action Required"
- Item name (e.g., USB Hub 4-Port)
- Current OnHand quantity (3)
- Configured ReorderPoint (5)

The alert fires only once per stockout event — the AlertSent flag ensures no duplicate emails until the item is restocked.

4.3 Reset AlertSent Flow — Logic

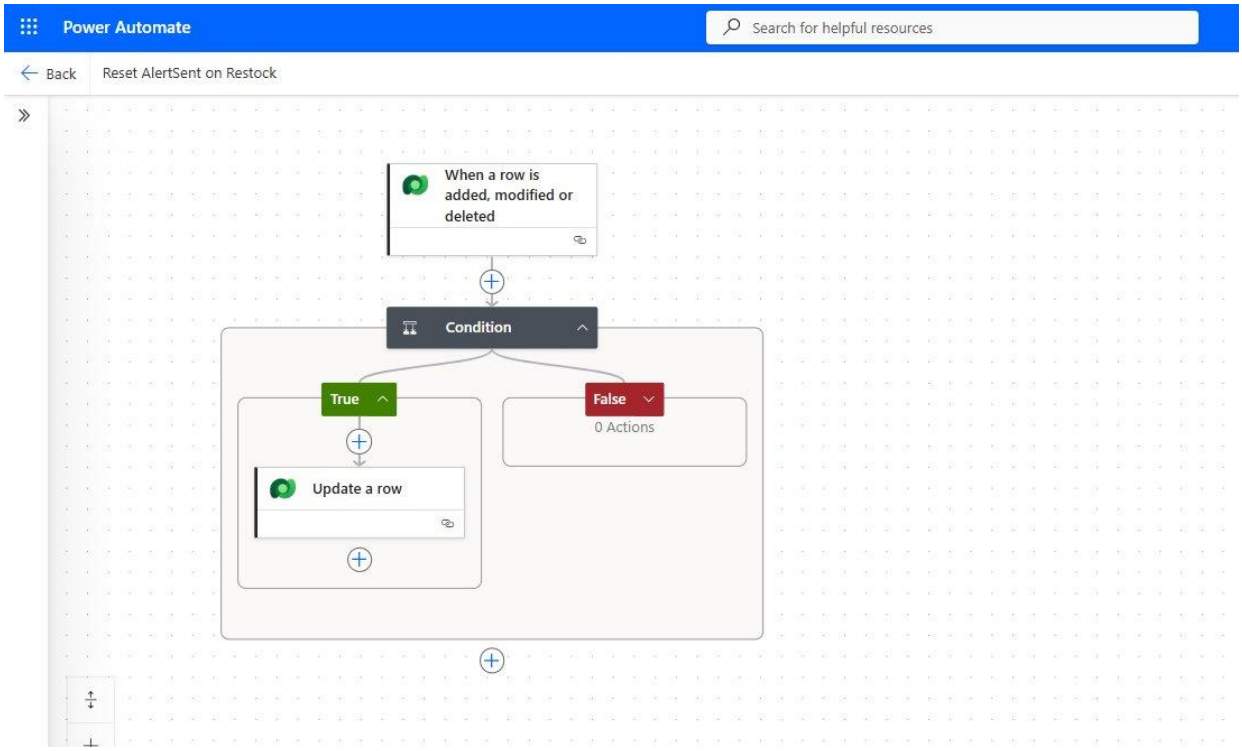


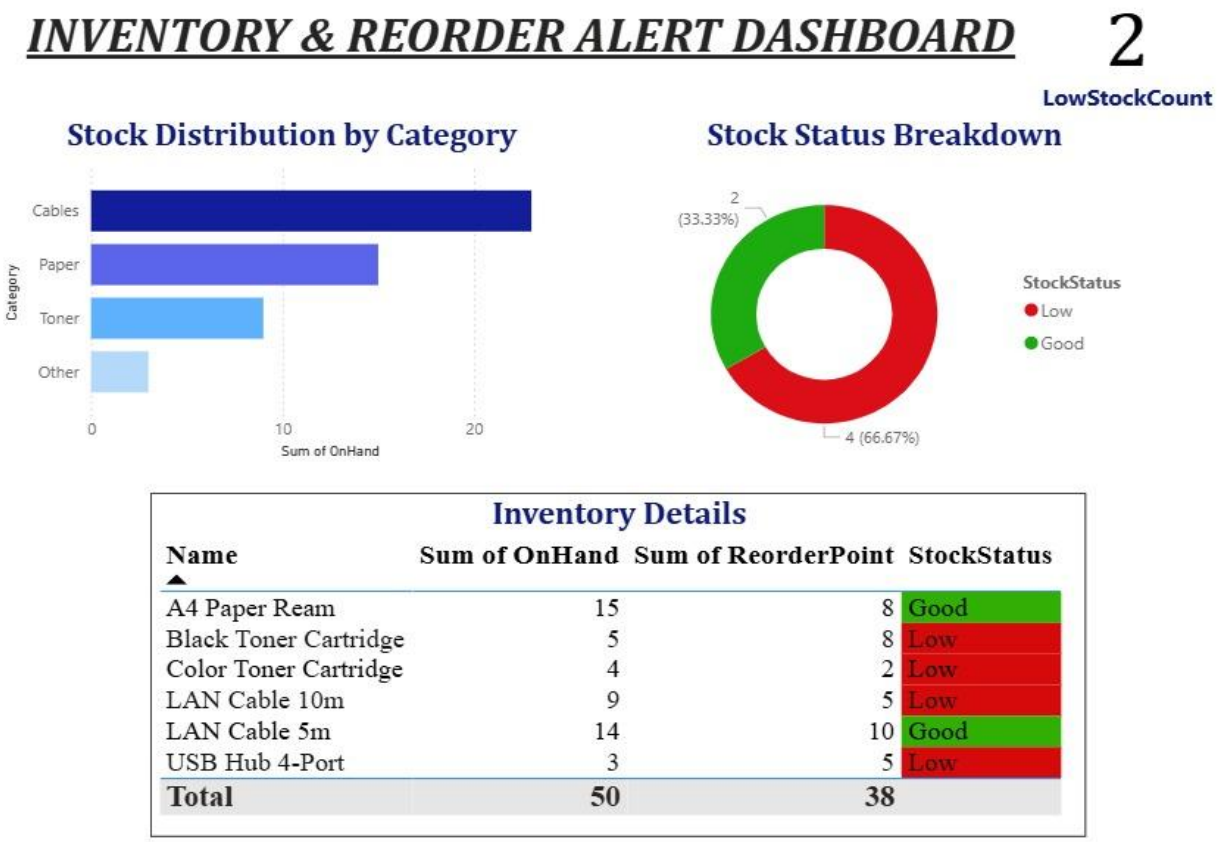
Figure 7: "Reset AlertSent on Restock" flow — triggered on row change, condition checks if restocked, updates AlertSent to No

The automated flow logic:

- Trigger: "When a row is added, modified or deleted" on the Items table
- Condition: checks if OnHand is now greater than or equal to ReorderPoint
- True branch: Updates the row — sets AlertSent = No, allowing future alerts to fire
- False branch: No action taken (item still below threshold)

5. Power BI — Inventory Dashboard

A compact Power BI report was built to give managers a visual overview of stock health across all categories.



Name	Sum of OnHand	Sum of ReorderPoint	StockStatus
A4 Paper Ream	15	8	Good
Black Toner Cartridge	5	8	Low
Color Toner Cartridge	4	2	Low
LAN Cable 10m	9	5	Low
LAN Cable 5m	14	10	Good
USB Hub 4-Port	3	5	Low
Total	**50**	**38**	

Figure 8: Power BI Inventory & Reorder Alert Dashboard — stock distribution, status breakdown and inventory details table

5.1 Dashboard Components

Visual	Description
LowStockCount KPI Card	Large number card showing count of items currently below reorder point (2 items)
Stock Distribution by Category	Horizontal bar chart showing total OnHand by category (Cables, Paper, Toner, Other)
Stock Status Breakdown	Donut chart showing proportion of items with Good vs Low stock status — 66.67% Low, 33.33% Good
Inventory Details Table	Full table with Name, Sum of OnHand, Sum of ReorderPoint, and color-coded StockStatus (green = Good, red = Low)

5.2 Key Insights from Dashboard

- Total OnHand across all items: 50 units | Total ReorderPoints sum: 38
- 4 out of 6 items (66.67%) are currently in Low stock status
- Cables category has the highest total OnHand (24 units combined)
- Toner category has critical low stock — both Black and Color Toner below reorder point

6. App Link

Power App: https://apps.powerapps.com/play/e/c5e4d9b6-d3f1-e14f-9623-8aea0f870907/a/cb80305e-2f8a-428a-93b4-de895ddefea5?tenantId=0c74de46-2a0e-4860-bda9-f34e07b931d4&hint=4af00068-efe6-45b7-862e-3beae3add0d9&sourcetime=1774786109090